

Class 10 Mathematics

ICSE and CBSE Board Syllabus

ICSE Syllabus

Theme 1: Commercial Mathematics

- Goods and Services Tax (GST): taxable amount, output/input tax, net GST, bill-based problems.
- Banking and shares & dividends: savings/current accounts, interest, recurring deposit, dividend, return on investment.

Theme 2: Algebra

- Linear inequations in one variable and their graphical representation on number line.
- Quadratic equations: factorisation, completing square, formula-based solutions, discriminant idea.
- Ratio and proportion; direct/inverse variation.
- Factorisation of polynomials including identities and remainder/factor ideas.
- Matrices: order, equality, addition/subtraction, scalar multiplication and simple multiplication.
- Arithmetic progression and geometric progression with n th term / sum applications.

Theme 3: Geometry

- Reflection and symmetry in coordinate geometry.
- Similar triangles and basic proportionality applications.
- Circle theorems: tangent properties, angles, chords, cyclic quadrilaterals.

Theme 4: Mensuration

- Cylinder, cone and sphere: CSA/TSA/volume.
- Combination solids and conversion/application problems.

Theme 5: Trigonometry

- Trigonometric ratios, complementary angles and identities.
- Heights and distances based on right triangles.

Theme 6: Statistics and Probability

- Mean, median, mode from grouped / ungrouped data.
- Histogram / ogive / cumulative frequency interpretation where applicable.
- Basic probability of simple events.

CBSE Syllabus

Unit 1: Number Systems & Algebra

- Real numbers: Euclid's division lemma, Fundamental Theorem of Arithmetic, irrational numbers, decimal expansion.
- Polynomials: zeros, relationship between zeros and coefficients, division algorithm.
- Pair of linear equations in two variables: graphical and algebraic solution.
- Quadratic equations and applications.
- Arithmetic progressions: n th term and sum of first n terms.

Unit 2: Coordinate Geometry

- Distance formula, section formula, area of triangle from coordinates.

Unit 3: Geometry

- Triangles: similarity, Pythagoras-based applications.
- Circles: tangent to a circle and number of tangents from an external point.

Unit 4: Trigonometry

- Introduction to trigonometric ratios, identities, and values of standard angles.
- Heights and distances.

Unit 5: Mensuration

- Surface areas and volumes of combinations of solids.

Unit 6: Statistics & Probability

- Mean of grouped data.
- Mode and median basics.
- Probability of simple events.